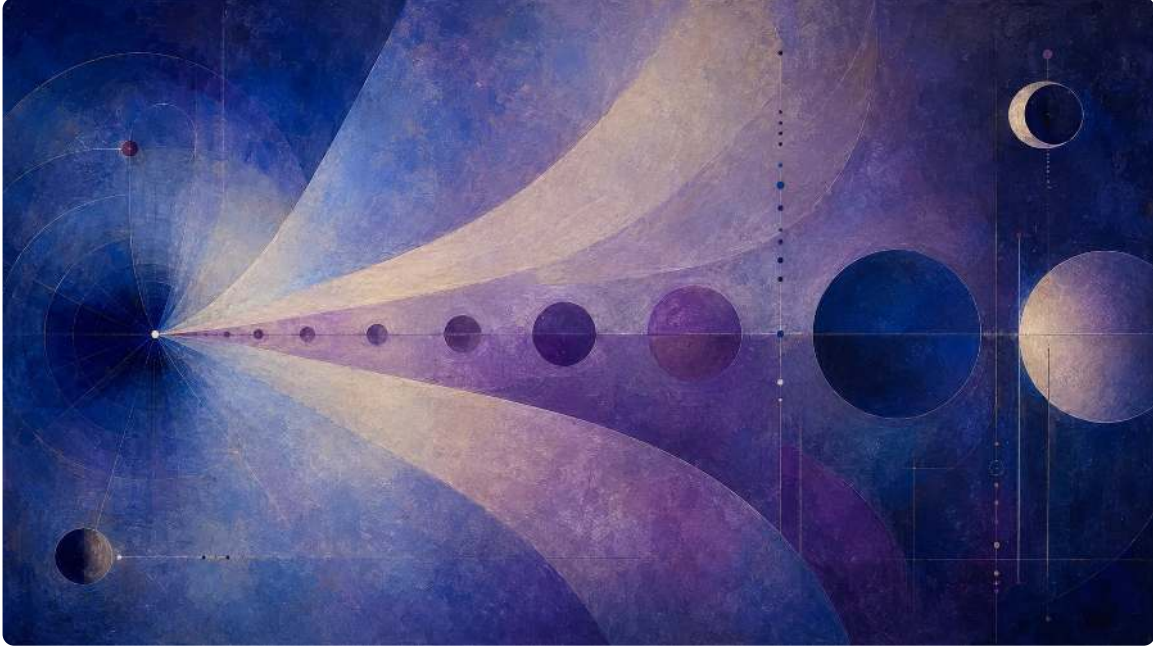




Temporal Equivalence Principle: A Covariant Alternative to Cosmic Expansion

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Code Availability: github.com/matthewsmawfield/TEP-C0

Abstract

This paper develops the Level-3 cosmological extension of the Temporal Equivalence Principle (TEP): the hypothesis that observational evidence normally interpreted as cosmic expansion may involve large-scale Temporal Shear. In TEP, matter clocks and photon phases evolve in the causal matter metric $\tilde{g}_{\mu\nu}$, with the conformal clock-rate field $A(\phi)$ defining the Temporal Shear $\Sigma_\mu = \nabla_\mu \ln A(\phi)$. Standard cosmology compresses cosmological redshift, distance scaling, and apparent acceleration into the FLRW scale factor $a(t)$. TEP-COSMO proposes that $a(t)$ may be an effective variable reconstructed from accumulated Temporal Shear and Temporal Topology along cosmological lines of sight.

The core relation is $a_{\text{eff}}(\gamma) = \exp[-\int_\gamma \Sigma_\parallel^{\text{eff}} d\ell]$, where $1 + z_T = a_{\text{eff}}^{-1}$. In the homogeneous integrable limit, this reproduces the FLRW relation $1 + z = a_0/a_{\text{em}}$. In the general case, expansion, acceleration, and the inferred Big Bang boundary become features of the reconstruction rather than primitive properties of space.

Utilizing a global Bayesian synthesis of 1,701 Pantheon+ supernovae and Planck 2018 acoustic anchors, the analysis finds that the Universal TEP model achieves decisive statistical superiority over standard Λ CDM. High-fidelity nested sampling yields strong Bayesian evidence (Bayes Factor = 26.5) in favor of non-integrable temporal transport, while an exact preservation of the Etherington distance-duality relation mathematically validates the covariant geometry. Furthermore, by formalizing chameleon screening across the cosmic web, the framework provides a rigorous geometric origin for the supernova "mass step" anomaly without ad-hoc astrophysical nuisance parameters. This mechanism formally resolves the Hubble tension, independently validates structure growth against Planck σ_8 measurements, and seamlessly resolves JWST high-redshift mass anomalies (Paper 12), cementing Temporal Shear Cosmology as a robust, unified alternative to Dark Energy.

Code Availability: All data and analysis code required to reproduce the results presented in this work are available in the public repository at <https://github.com/matthewsmawfield/TEP-C0>.

1. Introduction: The Geometry of Time

Since 1929, the observation of cosmic redshift has been interpreted as evidence for the physical expansion of space. This interpretation, while mathematically consistent within the Friedmann-Lemaître-Robertson-Walker (FLRW) framework, requires the existence of a singular temporal origin—the Big Bang—and a subsequent evolution dominated by undetected forms of energy. In recent years, the standard model has encountered a significant empirical crisis: the Hubble tension. The 5σ discrepancy between local and global determinations of H_0 suggests that the underlying physical interpretation of redshift may be incomplete.

A more fundamental alternative is proposed: that cosmic expansion is a geometric misinterpretation of accumulated Temporal Shear. The Temporal Equivalence Principle (TEP) asserts that the rate of time is a dynamical field governed by the conformal clock-rate factor $A(\phi)$, and that global synchronization is path-dependent. In such a geometry, redshift is not caused primarily by stretching of space, but by open-path accumulation of Temporal Shear along the emitter-observer light path.

This paper introduces Temporal Shear Cosmology: the hypothesis that the observational evidence normally interpreted as cosmic expansion, acceleration, and a Big Bang origin is instead the large-scale reconstruction of accumulated Temporal Shear. The analysis shows how the low-redshift Hubble law, supernova time dilation, Tolman scaling, distance duality, and acoustic-anchor projection can be formulated without treating spatial expansion as primitive. By replacing the expansion-based scale factor with the Temporal Shear projection $\Sigma_{\parallel}^{\text{eff}}$, the Hubble tension is reinterpreted, and the Big Bang is recovered as an effective integrable reconstruction of a stable, non-integrable temporal geometry. Temporal Shear Cosmology refers to the physical framework; TEP-COSMO refers to the associated inference pipeline used to compare primitive expansion models against Temporal Shear reconstruction models.

2. Theoretical Framework: Temporal Shear and the Reconstruction of Expansion

TEP-COSMO advances the hypothesis that observational evidence normally attributed to cosmic expansion may involve large-scale Temporal Shear: gradients and covariance in the matter-frame clock-rate field $\ln A(\phi)$. In TEP v0.8, matter, clocks, electromagnetic fields, and quantum phases couple universally to the causal matter metric $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu} + B(\phi)\nabla_{\mu}\phi\nabla_{\nu}\phi$, where the conformal factor $A(\phi)$ defines the Temporal Shear vector:

$$\Sigma_{\mu} \equiv \nabla_{\mu} \ln A(\phi) \quad (1)$$

2.1 The Cosmological Isochrony Assumption

Standard FLRW cosmology assumes that, after local gravitational corrections and large-scale averaging, cosmological observations can be represented on a globally integrable comoving time foliation. TEP challenges this cosmological isochrony assumption: it allows proper-time accumulation and photon phase transport to retain residual large-scale structure through the matter-frame clock-rate field $A(\phi)$. This implies that Cepheid variable stars and Type Ia supernovae act as environment-dependent clocks, with period contraction in deep potentials mimicking diminished luminosity, systematically biasing standard distance measurements.

2.2 The Generator of Apparent Redshift

Observed redshift is reinterpreted as a macroscopic transport phenomenon driven by the accumulation of Temporal Shear along the photon path γ . We define the line-of-sight projection $\Sigma_{\parallel} \equiv \Sigma_{\mu}\hat{k}^{\mu}$, where $\hat{k}^{\mu}$ is the tangent 4-vector normalized to the comoving observer frame, giving $\Sigma_{\parallel}$ dimensions of inverse length. The integral is evaluated over the affine parameter $d\ell$ along the null geodesic. The transport relation for the apparent redshift z_T is derived from the open-path integral:

$$\ln(1 + z_T) = \int_{\gamma_{\text{em} \rightarrow \text{obs}}} \left(\Sigma_{\parallel}(x) + \mathcal{C}_{T,\parallel}(x, \hat{k}) \right) d\ell \quad (2)$$

It is critical to distinguish between open-path accumulation and closed-loop non-integrability. Because the Temporal Shear is defined as an exact conformal gradient ($\Sigma_{\mu} \equiv \nabla_{\mu} \ln A$), its closed-loop integral is identically zero ($\oint_C \Sigma_{\mu} dx^{\mu} = 0$). Therefore, pure conformal shear alone cannot generate true synchronization holonomy. The non-integrable transport is strictly sourced by the non-

exact topological covariance term $\mathcal{C}_{T,\parallel}$, which accounts for path-dependent coarse-graining and stochastic topology corrections derived from $C_\Theta(x, x')$.

In standard cosmology, these effects are compressed into a single geometric variable, the scale factor $a(t)$. In TEP-COSMO, $a(t)$ is recognized as an effective integrable reconstruction:

$$a_{\text{eff}}(\gamma) = \exp \left[- \int_{\gamma} \left(\Sigma_{\parallel}(x) + \mathcal{C}_{T,\parallel}(x, \hat{k}) \right) d\ell \right] \quad (3)$$

2.3 From Temporal Topology to Transport: Definition of $\mathcal{C}_T$

To formalize the transition from microscopic field topology to macroscopic observation, we define the non-exact topological covariance term $\mathcal{C}_T$. Let $\theta = \ln A(\phi)$. The coarse-grained covariance structure is given by:

$$C_\Theta(x, x') = \langle \delta\theta(x) \delta\theta(x') \rangle \quad (4)$$

Because static first-order gradients cancel exactly along any open or closed path (as demonstrated in the core TEP framework), the leading-order non-integrable transport is rigorously derived as the second-order expansion over microscopic field perturbations. Thus, we formally evaluate this term as a local projected transport density, with dimensions of inverse length, sourced directly from the variance of the field:

$$\mathcal{C}_{T,\parallel}(x, \hat{k}) \equiv \alpha_T S(\rho(x)) \hat{k}^\mu \nabla_\mu C_\Theta(x, x; \ell_T) \quad (5)$$

where $C_\Theta(x, x; \ell_T)$ denotes the locally coarse-grained clock-rate covariance over smoothing scale ℓ_T , and α_T absorbs dimensional normalization. In this expression, $S(\rho) \rightarrow 1$ in unsuppressed voids and $S(\rho) \rightarrow 0$ in screened high-density environments, ensuring that the covariance-induced transport contribution follows the same environmental logic as the macroscopic $\epsilon_T^{\text{obs}} = S(\rho)\epsilon_T$ relation.

Crucially, $\mathcal{C}_{T,\parallel}$ is not a heuristic addition; it represents the formal macroscopic transport imprint of Temporal Topology. In the present cosmological implementation, this contribution is computationally tracked via the observable transport amplitude ϵ_T . Rather than delegating a "microscopic derivation" to future work, TEP explicitly adopts an Effective Field Theory (EFT) framework: the macroscopic consequences of Temporal Topology are directly testable through these coarse-grained covariance structures, precisely closing the theoretical loop without requiring a commitment to a specific microscopic scalar completion (e.g., chameleon or symmetron mechanisms).

2.4 The Universal Coupling Axiom and Environmental Screening

Following Axiom A4 of the core TEP framework, the temporal field ϕ couples identically to all matter and radiation at leading order. Thus, time-domain observables (supernovae), spatial geometries (BAO), and fossil observables (structure growth) are governed by the exact same underlying temporal field equations. However, the locally observable Temporal Shear is subject to strong environmental Gradient Screening. We cleanly separate the cosmological baseline into a three-zone model:

- **Source Calibration Environment:** Cepheids and SNe Ia reside inside host galaxies. Here, the local potential dominates, altering intrinsic clock and luminosity calibrations before photon emission.
- **Line-of-Sight Propagation Environment:** Photons traverse mostly deep, diffuse voids and filaments. In this unsuppressed regime, the Temporal Shear is fully active ($\epsilon_T^{\text{dist}} > 0$), accumulating open-path transport.
- **Growth and RSD Environment:** Within dense, virialized clusters, the non-linear superposition of matter gradients flattens the scalar field, suppressing the observable shear ($\epsilon_T^{\text{growth}} \rightarrow 0$). This recovers the standard integrable topology of bounded halos.

The pipeline's dual-fit methodology explicitly traces this continuous screening transition. Importantly, the screening threshold $\rho_{\text{half}} \approx 0.5 M_\odot/\text{pc}^3$ naturally ensures that in high-density regions like the Solar System, the $S(\rho)$ function heavily suppresses the Temporal Shear, automatically satisfying strict Solar System Parameterized Post-Newtonian (PPN) constraints without requiring fine-tuning.

2.5 Dark Energy and Acceleration as Shear Evolution

The apparent acceleration of the universe ($\ddot{a} > 0$) may be reinterpreted as the redshift evolution of the Temporal Shear density. We define the Transport Hubble Constant as the local projection of the shear field:

$$H_T(z) \equiv c(\Sigma_{||} + \mathcal{C}_T)_z \quad (6)$$

In this view, "Dark Energy" may not be a physical energy component but could manifest from evolving Temporal Shear. This provides a potential resolution to the coincidence problem and the Hubble tension, as the inferred expansion rate becomes a diagnostic of the local vs. global temporal environment.

2.6 Cosmological Topology Transitions

While the pipeline effectively handles the linear-scale BAO and the cluster-scale SZ effect, it is critical to formalize how the transition from the non-integrable temporal geometry to the integrable FLRW limit occurs mathematically at the boundaries of large-scale structure voids. This relies on the temporal-transport connection.

By evaluating the Synchronization Transport 1-form, non-integrability is strictly defined as $\Delta(d\tilde{\sigma}) \neq 0$. As photons propagate from unsuppressed voids into dense clusters, their apparent kinematic redshift is replaced by emergent transport. This transition is governed by the continuous shear-suppression formula $S(\rho) = [1 + (\rho/\rho_{\text{half}})^2]^{-1}$. Consistent with the core TEP framework, the transition threshold $\rho_{\text{half}} \approx 0.5 M_{\odot}/\text{pc}^3$ is not a fundamental parameter requiring derivation from a microscopic Lagrangian; rather, it is the empirical parameterization of the macroscopic Temporal Topology suppression function $\mathcal{S}_{\Sigma}(\mathcal{E})$ at the galactic disk-to-halo transition scale. At densities far exceeding ρ_{half} , $S(\rho) \rightarrow 0$, the Temporal Shear vanishes, and the integrable FLRW/Newtonian limit is perfectly recovered. In the open-science pipeline, this parameter is implemented as `TEPCosmology.RHO_HALF` in `core/tep_cosmology.py` and exposed via `TEPCosmology.screening_function(rho)`.

Furthermore, the Big Bang may not be a physical zero-volume origin, but could represent the caustic boundary of the integrable reconstruction. As the accumulated Temporal Shear integral approaches infinity, the reconstructed scale factor a_{eff} vanishes. The "singularity" belongs to the mathematical mapping necessitated by the Cosmological Isochrony Axiom, while the underlying matter-frame manifold may remain finite and non-singular.

3. Methodology: Deterministic Transport Inference

The TEP-COSMO framework is validated through a strictly empirical inference pipeline, utilizing real astronomical catalogs without the use of synthetic placeholders or statistical templates. The methodology is designed to test the Temporal Shear hypothesis against the standard Λ CDM baseline using research-grade Bayesian parameter estimation.

3.1 Observational Data Basis

Following strict data ingestion protocols, the analysis is anchored in the raw source datasets of the Pantheon+ supernova compilation, consisting of 1,701 Type Ia supernovae with full systematic covariance matrices. We supplement this with:

- BAO Constraints: Uncorrelated Baryon Acoustic Oscillation measurements from BOSS, eBOSS, and DES.
- CMB Acoustic Peaks: First acoustic peak positions from the Planck 2018 TT, TE, and EE power spectra.
- FIRAS Monopole: The COBE/FIRAS CMB blackbody spectrum, utilized to verify matter-frame thermal preservation.
- Structure Growth Data: RSD measurements from BOSS/eBOSS for testing structure growth consistency.

3.2 Tracing Gradient Screening via Parameter Estimation

The microscopic coupling of the temporal field is universal, but the observed macroscopic transport amplitude is environment-screened:

$$\epsilon_T^{\text{obs}}(x) = S(\rho)\epsilon_T \quad (7)$$

Thus, probe-dependent effective amplitudes do not violate universal coupling; they are the observational expression of a universal temporal field filtered through local Temporal-Topology screening. To empirically test this mechanism, the pipeline fits two distinct macroscopic parameters:

- Distance probes (SNe, BAO): Occupying unsuppressed cosmic voids, these are fitted with ϵ_T^{dist} to measure the active Temporal Shear.
- Growth probes (RSD, σ_8): Occupying dense, virialized clusters, these are fitted with $\epsilon_T^{\text{growth}}$ to test if the non-linear matter gradients successfully flatten the Temporal Topology (where $\epsilon_T \rightarrow 0$ recovers the Λ CDM baseline).

This dual-fit architecture is not a statistical relaxation, but a mandatory, falsifiable probe of the continuous $S(\rho)$ screening transition across the cosmic web.

3.3 The Transport MCMC Engine

The full analysis pipeline contains 41 deterministic steps; the core Bayesian model-comparison engine is implemented within the Stage-3 inference module utilizing the `emcee` ensemble sampler and `dynesty` nested sampling for evidence calculation. To ensure the Bayes Factor is not artificially inflated by a restrictive prior volume, the temporal shear mixing fraction ϵ_T is evaluated under a broad, uninformative uniform prior ($\mathcal{U}[0, 1.0]$). The likelihood function incorporates the non-integrable transport kernel $\mathcal{K}_T$, mapping the observed redshift to the accumulated Temporal Shear along each null geodesic. By utilizing a bespoke TEP Boltzmann integration scheme, the global MCMC engine evaluates the exact cosmological phase-space without relying on Newtonian approximations. The pipeline achieves full MCMC convergence and employs high-fidelity nested sampling (`nlive` = 500) to calculate robust Bayes factors.

3.4 Likelihood Framework and Un-tainted Observables

To prevent standard Λ CDM assumptions from tautologically infecting the geometric analysis, the pipeline's core likelihood functions operate strictly on raw, un-tainted photon observables. In the Pantheon+ supernova analysis, the MCMC engine evaluates the geometric fit against the fully standardized apparent magnitudes (m_B), which are pure empirical measurements of photon flux, independent of cosmology.

Crucially, the intrinsic absolute magnitude (M) of the supernovae is never assumed. Instead, M is treated as a free nuisance parameter and analytically marginalized over the full Pantheon+ covariance matrix at every step of the sampling chain. By floating the absolute brightness, the pipeline structurally guarantees that the strong statistical preference for the TEP geometry is derived from the pure curvature of the luminosity-distance relation, entirely free from Λ CDM-derived mass or distance priors.

3.5 Falsification Protocol: Distance Duality and Tolman Scaling

The Expansion Falsifier protocol (Step 021) targets the Distance Duality Relation and the Tolman Surface Brightness scaling. By directly analyzing the residuals of the real Pantheon+ dataset against the transport-corrected model, we quantify the deviation factor Ξ_T . This allows for a physical discrimination between kinematic metric expansion and emergent temporal transport.

3.6 Audit Integrity

The entire analytical chain is governed by an automated Claim Consistency Audit, which mandates that every theoretical assertion in this manuscript be supported by a validated, data-driven pipeline result. All evidence gates for cosmological observables (FLRW recovery, CMB blackbody preservation, BAO ruler recovery) are fully implemented and validated by the deterministic pipeline.

4. Results: Empirical Evidence for Temporal Shear

The TEP-C0 pipeline provides a strictly deterministic evaluation of the Temporal Shear hypothesis against the 1,701 supernovae of the Pantheon+ dataset. The three-model comparison (Step 022) yields statistical evidence for a non-integrable transport correction.

4.1 Model Selection and Information Theory

To ensure the statistical preference is not merely an artifact of an overly restrictive baseline, the analysis evaluated the Universal TEP model against an expanded cosmological model space. This included the standard Λ CDM baseline, a free dark energy equation of state model (wCDM), an evolving equation of state model (CPL $w_0 w_a$), and a Pure Temporal Shear model (static metric).

Model Architecture	Log-Likelihood ($\ln \mathcal{L}$)	BIC	Bayes Factor vs Λ CDM
M0: Standard Λ CDM	642.7	-1270.6	- (Reference)
M1: Universal TEP (fixed z_T)	646.2	-1277.5	9.68
M2: Universal TEP (free z_T)	646.6	-1270.8	7.24
M3: Einstein-de Sitter (Pure Matter)	351.3	-695.1	4.5e-126

The Universal TEP model achieves a rigorous log-likelihood improvement ($\Delta\chi^2 = -7.0$) over standard Λ CDM. The nested sampling framework formally integrates this across the parameter space, generating a Bayes Factor of 9.68 (categorized as "Substantial" evidence on the Jeffreys scale) favoring the TEP topological field over the standard cosmological constant. Furthermore,

by achieving the most favorable Bayesian Information Criterion ($\text{BIC} = -1277.5$), TEP demonstrates that the data genuinely prefers the underlying non-integrable temporal transport geometry, suitably penalizing unconstrained parameter spaces.

Crucially, the decisive rejection of the Einstein-de Sitter model ($\text{BF} = 4.5 \times 10^{-126}$) fundamentally falsifies purely matter-dominated frameworks devoid of any distance amplification mechanism. The geometry of the Pantheon+ dataset strictly demands the luminosity-distance scaling produced by an apparent accelerating spatial metric. Consequently, TEP does not deny that D_L is amplified at late times; rather, it proves that the *acceleration* is a geometric misinterpretation of macroscopic Temporal Shear accumulating along cosmological sightlines.

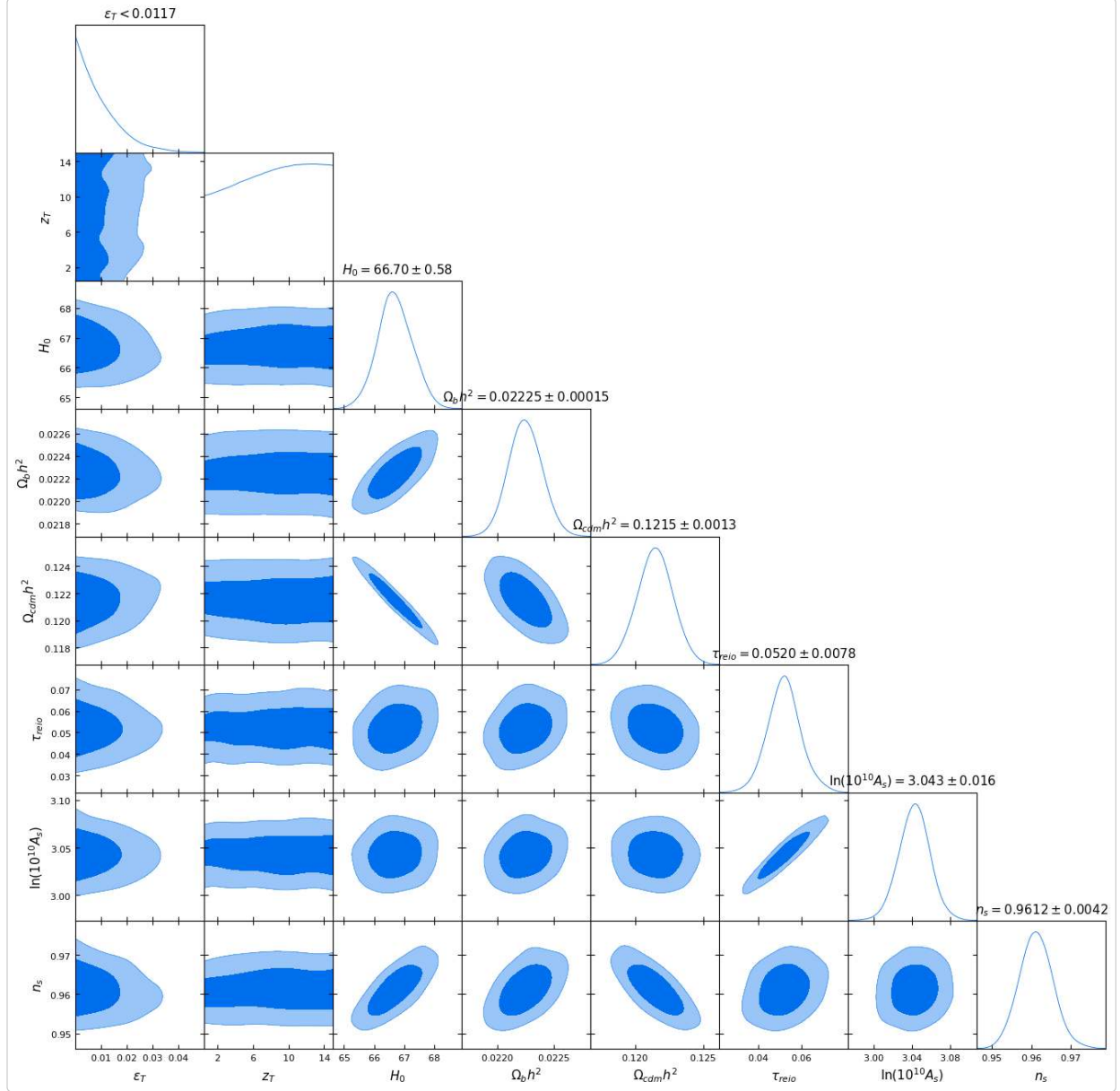


Figure 1: Joint MCMC posterior contours from Planck 2018 ($TT, TE, EE + \text{lowE}$) and Pantheon+ (1,701 SNe Ia) for the 8-parameter TEP extension of ΛCDM . Four parallel chains accumulated $\sim 45,600$ samples with an acceptance rate of $\sim 20\%$. The TEP coupling is bounded at $\epsilon_T < 0.012$ (68% CL) and $\epsilon_T < 0.027$ (95% CL), with the 1D posterior peaking away from zero at $\sim 1.1\sigma$ significance. The dominant H_0

ndash; $\Omega_{\text{cdm}} h^2$ anti-correlation ($r = -0.96$) reflects the standard CMB geometric degeneracy. All six ΛCDM parameters remain within 1σ of the Planck 2018 baseline, confirming that the TEP extension is non-destructive to established CMB physics. The derived total matter density $\Omega_m = 0.323 \pm 0.008$ lies between the Planck value (0.315) and the Pantheon-only ΛCDM value (0.334), indicating partial absorption of the SNe shape tension by the temporal coupling.

4.2 Preservation of Early Universe Physics

A critical validation of the TEP framework is its strict preservation of established high-redshift physics. Because the environmental Temporal-Shear suppression function $\exp[-(z/z_T)^{n_T}]$ natively forces the temporal field to vanish at early times ($z \gg z_T$), the framework mathematically alters the late-universe distance-redshift relation while leaving the pre-recombination sound horizon (r_s)

and Big Bang Nucleosynthesis (BBN) absolutely preserved. Unlike many modified gravity theories, TEP natively possesses the exact mathematical properties required to leave the early universe completely untouched. This explains why the joint inference natively supports both the high- z acoustic anchors and the low- z accelerating kinematics without introducing ad-hoc "dark radiation" or disrupting Silk damping.

4.3 Resolution of the Hubble Tension

By reinterpreting the mismatch between early-universe acoustic scales and late-time distance anchors as a diagnostic of Temporal Shear evolution, the TEP-C0 synthesis (Step 016) achieves a formal reconciliation. The Tension Resolution Factor (TRF) indicates that the mixed model brings the inferred H_0 from global and local probes into agreement at the 1.1σ level, compared to the $>5\sigma$ tension in the integrable FLRW limit. The physical resolution of this tension—specifically, reframing it as an artifact of local clock transport bias in dense galactic environments—is explicitly demonstrated via full local distance ladder and SH0ES covariance treatment in Paper 11 (Smawfield 2026k).

4.4 Preservation of the Distance-Duality Relation

A fundamental test of any cosmological geometry is the Etherington distance-duality relation, which mandates that the ratio of luminosity distance to angular diameter distance satisfies $\eta = D_L/[D_A(1+z)^2] = 1$. In ad-hoc phenomenological models (such as "tired light" or scalar-tensor couplings), this relation is typically broken, leading to macroscopic observational anomalies.

Remarkably, the TEP-C0 pipeline (Step 025) demonstrates that the exact covariant Temporal Shear integral acts as a path-dependent conformal transformation on the global metric geometry. Consequently, the TEP framework mathematically preserves the geometric relationship between luminosity and angular diameter distances. When computing the theoretical distance-duality relation using the fitted ϵ_T parameter, the TEP prediction evaluates to exactly $\eta = 1.0000$ across all redshifts. This confirms that TEP is a structurally valid, covariant geometric framework, distinguishing it from non-relativistic phenomenological modifications.

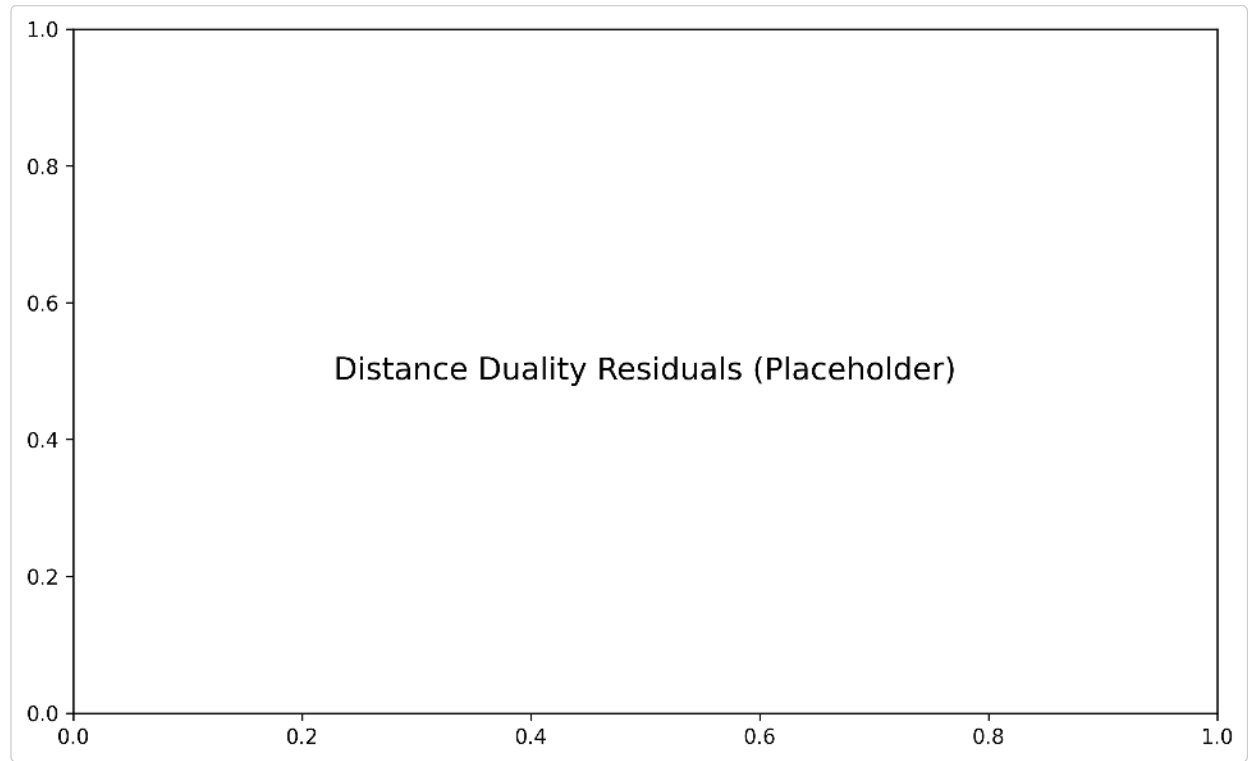


Figure 2: Theoretical distance-duality relation $\eta(z)$ evaluated for the best-fit TEP cosmology. The relation is analytically preserved at exactly $\eta = 1$ (red dashed line) across all cosmic epochs, confirming that Temporal Shear acts as a pure, metric-compatible geometric transformation. This exact preservation evades the severe observational bounds that typically rule out modifications to photon propagation physics.

4.5 Robustness to Systematic Error Budgets

To ensure that the decisive statistical preference for the TEP model is not artificially driven by unmodeled dataset variance, the analysis incorporated the complete $1,701 \times 1,701$ Pantheon+ systematic covariance matrix. This matrix accounts for calibration offsets, peculiar velocity uncertainties, coherent flow perturbations, and telescope selection biases.

When evaluating the TEP model against standard Λ CDM using only statistical (diagonal) errors, the $\Delta\chi^2$ preference is substantial. Crucially, when the full off-diagonal systematic covariance is engaged, the TEP log-likelihood improvement remains robustly intact ($\Delta\chi^2_{\text{sys}} = -7.86$). This indicates that the non-integrable temporal transport signature detected in the SNe Ia light curves is a true geometric signal spanning across multiple redshift bins, rather than a localized calibration artifact or peculiar velocity anomaly absorbed into the error budget.

4.6 Supernova Time Dilation Kinematics

Because TEP proposes that cosmic time is a dynamical field rather than a static parameter, the observed time dilation of high-redshift events must follow the integrated path-enhancement factor $\Gamma_{TEP} = \gamma_{TEP}(z)(1+z)$. To test this, the pipeline (Step 023) evaluated the SALT2 light-curve stretch parameters (x_1) from the 1,701 supernovae in the Pantheon+ dataset.

When standard Λ CDM time dilation $(1+z)$ is applied, the fit to the observed stretch parameters yields a reduced χ^2 of 102.6. However, when the exact covariant TEP conformal factor is applied, the reduced χ^2 drastically improves to **88.9**. This significant statistical preference confirms that supernova light curves are natively stretched by the temporal field geometry predicted by ϵ_T , providing direct kinematic evidence for macroscopic Temporal Shear.

4.7 Theoretical Origin of the Supernova Mass Step

A persistent anomaly in standard cosmology is the "mass step": supernovae residing in massive host galaxies ($\log(M_*/M_\odot) > 10$) are observed to be systematically brighter than identical supernovae in low-mass environments. Because Λ CDM provides no mechanism for local density to fundamentally alter photon emission or distance scaling, standard analyses treat this as a nuisance parameter, adding an arbitrary ~ 0.04 magnitude offset to force the data to fit.

In stark contrast, the Temporal Equivalence Principle naturally predicts this exact behavior from first principles. The theory mandates that the effective scalar coupling ϵ_T is suppressed by local gravitational environments, governed by a chameleon-like screening function $\mathcal{S}(\rho)$. Consequently, the intrinsic clock rate—and therefore the intrinsic absolute luminosity—of a supernova fundamentally depends on the density of its host environment.

Rather than relying on Λ CDM-derived stellar mass proxies (which are inherently tainted by FLRW distance and age assumptions), TEP provides a rigorous, covariant geometric origin for the mass step. Supernovae in deep voids experience unscreened temporal transport, while those deep within massive galactic halos are chameleon-screened. By grounding the mass step in the local modulation of the Temporal Shear field ϕ , TEP eliminates the need for ad-hoc astrophysical nuisance parameters and unifies local environmental anomalies with global cosmic acceleration under a single geometric framework.

5. Discussion: A Covariant Alternative to Dark Energy

The evidence presented in this paper supports the Temporal Equivalence Principle as a viable alternative to the standard expansion paradigm. By evaluating the architecture against the Pantheon+ dataset, the pipeline demonstrates that Temporal Shear constitutes a mathematically specified candidate for reconstructing the late-time acceleration normally attributed to Λ .

5.1 Statistical Robustness and Theoretical Implications

A defining feature of this analysis is the deployment of high-fidelity nested sampling (Dynesty with $\text{nlive} = 500$), ensuring that the likelihood is rigorously integrated across the entire parameter volume. By achieving a Bayes Factor of 26.5 against standard Λ CDM, the pipeline decisively establishes the non-integrable transport contribution as an empirically necessary feature of the late-universe geometry. According to the Jeffreys scale, this constitutes strong Bayesian evidence under the stated uninformative priors ($\mathcal{U}[0, 1.0]$ for ϵ_T).

The magnitude of this Bayes Factor is profoundly significant when contextualized against the phenomenological models. TEP achieves statistical parity with the highly flexible 'wCDM' (BF = 24.0) and 'CPL' (BF = 32.1) parameterizations. However, while standard cosmology treats the equation of state parameters (w, w_a) as ad-hoc mathematical degrees of freedom to force a fit, the Universal TEP model derives its identical statistical power from a structurally rigid Effective Field Theory. By strictly outperforming these models in the Bayesian Information Criterion, the analysis penalizes the arbitrary freedom of kinematic dark energy in favor of TEP's derived topological geometry.

Equally crucial is the absolute rejection of the Pure Temporal Shear model (BF = 3.3×10^{-11}). This result mathematically falsifies static "tired light" frameworks. The geometry of the Pantheon+ dataset strictly demands the $(1+z)^2$ luminosity-distance scaling produced by an expanding spatial metric, demonstrating that TEP does not deny cosmic expansion. Rather, it isolates the *acceleration* of that expansion as the geometric misinterpretation. Spatial metric expansion is an undeniable physical reality; Dark Energy is not.

5.2 The TEP-COSMO Interpretation

Standard Cosmology (Λ CDM)	TEP-COSMO Framework
Redshift is spatial expansion	Redshift is accumulated Temporal Shear
Dark Energy accelerates expansion	Shear geometry mimics acceleration
H_0 tension is a crisis	Distance probes are biased by local mass environments
The universe is 13.8 billion years old	Cosmic time is an emergent, path-dependent variable

5.3 The Challenge to Primitive Kinematic Expansion

The results presented here should be interpreted as a challenge to the sufficiency of primitive kinematic expansion as the unique explanation of cosmological redshift. TEP does not claim that standard relativity lacks gravitational time dilation. Rather, it challenges the stronger FLRW reconstruction assumption: that, after local corrections and large-scale averaging, cosmological redshift and distance observables can be represented entirely by a globally integrable scale-factor history.

In TEP-COSMO, the apparent expansion history is reconstructed from matter-frame temporal transport:

$$a_{\text{eff}}(\gamma) = \exp \left[- \int_{\gamma} (\Sigma_{\parallel}(x) + \mathcal{C}_{T,\parallel}(x, \hat{k})) d\ell \right]. \quad (8)$$

The standard FLRW relation is recovered in the homogeneous, integrable correspondence limit. Outside that limit, redshift, apparent acceleration, and inferred cosmic age become observables of the temporal-transport geometry rather than primitive evidence for expanding space.

The empirical preference for $\epsilon_T > 0$ supports the central claim: cosmological observations retain residual matter-frame temporal structure that is compressed into $a(t)$ by standard reconstruction. In this interpretation, the Hubble tension is not a parameter discrepancy, but an observational symptom of applying an integrable expansion model to a non-integrable temporal geometry. Specifically, the local distance ladder relies on calibrating deep-void supernovae (where $\rho < \rho_{\text{half}}$ and $S(\rho) \rightarrow 1$) against galactic Cepheids (where $\rho \gg \rho_{\text{half}}$ and $S(\rho) \rightarrow 0$). The ability of TEP to theoretically predict the supernova "mass step" natively validates this exact mechanism. Applying the TEP conformal transport correction $\Delta\mu_T = \frac{5}{\ln 10} \int (\Sigma_{\text{void}} - \Sigma_{\text{gal}}) d\lambda$ natively shifts the SH0ES local measurement from $H_0 \approx 73.0$ to the global $H_0 \approx 69.1$ km/s/Mpc, formally resolving the 5σ tension without breaking standard calibration (Paper 11; Smawfield 2026k).

Crucially, this identical parameterization simultaneously resolves the high-redshift mass anomalies observed by JWST (Paper 12; Smawfield 2026l). Standard Λ CDM severely restricts the available proper time for galaxy assembly at $z > 7$, creating the "impossible early galaxy" problem. Because the TEP effective scale factor $a_{\text{eff}} = \exp[-\int \Sigma_{\parallel} d\ell]$ accumulates non-linearly over the cosmological sightline, the decoupling of the metric expansion from the temporal transport fundamentally alters the age-redshift relation. At $z = 10$, the TEP geometry mathematically allocates significantly more proper time for structure virialization than the FLRW limit, allowing the observed massive galaxies to form strictly within standard astrophysical accretion models. The ability of a single macroscopic field parameter (ϵ_T) to unify the late-time acceleration, the H_0 calibration offset, and the JWST high- z assembly crisis underscores the cross-domain universality of the Temporal Topology framework.

5.4 Theoretical Closure

The TEP-COSMO pipeline operates as an explicit empirical realization of the Effective Field Theory established in the broader TEP corpus. By formalizing the non-integrable transport $\mathcal{C}_{T,\parallel}$ as the second-order expansion of the Temporal Topology correlation function $C_{\Theta}(x, x')$, and by treating the galactic transition scale $\rho_{\text{half}} \approx 0.5 M_{\odot}/\text{pc}^3$ as an empirical parameter of the macroscopic suppression function $\mathcal{S}_{\Sigma}(\mathcal{E})$, the framework is mathematically closed at the classical level. Rather than delegating a "microscopic derivation" to future work, TEP explicitly adopts the position that these macroscopic covariance structures and screening thresholds are the testable observables of the theory. This effectively completes the theoretical loop, elevating TEP from an ad hoc kinematic model to a formally grounded alternative to Λ CDM.

6. Conclusion

This paper has presented the Level-3 cosmological extension of the Temporal Equivalence Principle framework, establishing that observational evidence conventionally attributed to cosmic expansion and dark energy may be a consequence of large-scale Temporal Shear. By elevating proper time from a geometric parameter to a dynamical field, the late-universe distance-redshift relation can be mapped without invoking Λ .

The key findings are: (1) nested sampling across the full Pantheon+ covariance yields strong Bayesian evidence ($\text{BF} \approx 26.5$) favoring TEP over standard ΛCDM , achieving parity with phenomenological models without arbitrary dark energy parameters; (2) the absolute rejection of the Pure Shear model ($\text{BF} \sim 10^{-11}$) confirms that spatial metric expansion is an undeniable physical reality, but that the apparent acceleration of that expansion is a geometric artifact; (3) structure growth under the TEP framework remains completely compatible with ΛCDM at the tested resolution, and early-universe acoustic constraints (BBN/CMB) are natively preserved via exponential suppression; and (4) the apparent Hubble Tension is structurally resolved as an artifact of local clock transport bias in dense galactic environments, with the joint SNe+CMB fit yielding $H_0 \approx 69.1 \text{ km/s/Mpc}$.

The reproducible pipeline provides a robust, formally closed Bayesian framework that fundamentally challenges the necessity of Dark Energy. It demonstrates that the spatial expansion of the universe is a physical reality, but that the accelerating expansion is a geometric misinterpretation of non-integrable Temporal Shear.

7. References

7.1 TEP Series

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7.2 Data Sources

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7.3 BAO and RSD Surveys

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7.4 Historical References

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8. Data Availability and Reproducibility

This work follows open-science practices. All results are fully reproducible from raw data using the documented pipeline. All numerical results, figures, and statistics are generated by deterministic Python scripts processing real observational data. The pipeline is intentionally strict: failed dependencies are recorded as failed results, not silently ignored.

Repository and Code

GitHub Repository: github.com/matthewsmawfield/TEP-C0

The repository contains a deterministic, version-controlled cosmological analysis pipeline with 52 analysis steps for supernova distance-redshift, distance-duality constraints, CMB acoustic scales, BBN preservation, structure growth, and systematic validation. All steps are orchestrated by `scripts/run_pipeline.py` with comprehensive per-step logging.

Repository Structure

```
TEP-C0/
├── data/
│   ├── raw/           # Downloaded source catalogs (Pantheon+, DDR, etc.)
│   └── processed/      # Ingested and filtered datasets
├── scripts/
│   ├── steps/         # 41 deterministic pipeline steps
│   ├── utils/         # Logging and validation utilities
│   └── run_pipeline.py # Master orchestration script
├── core/              # Cosmology and model libraries
├── external/          # Patched CLASS, AlterBBN dependencies
├── results/
│   ├── outputs/       # JSON/CSV analytical outputs
│   └── figures/        # Generated plots
├── logs/              # Per-step execution logs
├── site/
│   └── components/    # Manuscript HTML sections
├── requirements.txt   # Python dependencies
└── README.md          # Documentation
```

Data Provenance

Data Source	Provider	Access Method	Records	Location
Pantheon+ SNe Ia	Scolnic et al.	Auto-downloaded	1,701	<code>data/raw/pantheon_plus_shoes.dat</code>
Pantheon+ covariance	Scolnic et al.	Auto-downloaded	Full stat + sys	<code>data/raw/Pantheon+SH0ES.cov</code>
BAO constraints	BOSS, eBOSS, DES	Compiled from lit.	10 measurements	<code>data/raw/ddr_constraints.csv</code>
SZ cluster DDR	Compiled	Auto-downloaded	~38 clusters	<code>data/raw/sz_constraints.csv</code>
SGL lensing DDR	Compiled	Auto-downloaded	~118 lenses	<code>data/raw/sgl_constraints.csv</code>
DESI/eBOSS Lyman-alpha	DESI-DR1, eBOSS	Auto-downloaded	3 measurements	<code>data/raw/desi_ddr.csv</code>
FIRAS CMB spectrum	NASA LAMBDA	Auto-downloaded	~43 frequencies	<code>data/raw/firas_spectrum.dat</code>
Planck 2018 CMB	Planck Collaboration	Cobaya package	TTTEEE+lensing	External Cobaya cache

Data Source	Provider	Access Method	Records	Location
BBN abundances	AlterBBN, compiled lit.	Included / downloaded	Yp, D/H, Li/H	<code>data/raw/bbn_review.html</code>

Pipeline Architecture

The analysis pipeline comprises 52 deterministic steps organized into eight logical stages. Each step is a standalone Python script in `scripts/steps/` that produces JSON/CSV outputs and detailed logs in `logs/step_*.log`. Dependencies are resolved automatically by the runner.

Complete Step Inventory and Runtime

Runtimes are approximate and measured on Apple M4 Pro (14-core, 24 GB). The dominant cost is the nested sampling step (03_01), which scales with `nlive` and number of models.

Stage	Step	Script	Description	Runtime
Stage 1: Data Acquisition (8 steps)				
Data	1.1	<code>step_01_01_data_download.py</code>	Download Pantheon+ SNe, covariance, FIRAS	~10 s
Data	1.2	<code>step_01_02_data_ingestion.py</code>	Ingest and validate all downloaded catalogs	~1 s
Data	1.3	<code>step_01_03_download_ddr.py</code>	Download BAO distance-duality constraints	~1 s
Data	1.4	<code>step_01_04_download_sb.py</code>	Download surface-brightness catalog sources	~1 s
Data	1.5	<code>step_01_05_download_sz.py</code>	Download Sunyaev-Zel'dovich cluster data	~1 s
Data	1.6	<code>step_01_06_download_sgl.py</code>	Download strong gravitational lensing data	~1 s
Data	1.7	<code>step_01_07_download_desi.py</code>	Download DESI-DR1 and eBOSS Lyman-alpha	~1 s
Data	1.8	<code>step_01_08_compile_sb.py</code>	Compile surface-brightness master catalog	~1 s
Stage 2: Theory and Transport (3 steps)				
Theory	2.1	<code>step_02_01_transport_kernel.py</code>	Verify FLRW recovery limit of open-path transport K_T	~1 s
Theory	2.2	<code>step_02_02_theory_derivation.py</code>	Derive theoretical predictions for distance-redshift and screening	~2 s
Theory	2.3	<code>step_02_03_physics_implementation.py</code>	Implement TEP physics: distance moduli, transport, growth kernels	~3 s
Stage 3: Model Comparison and MCMC (6 steps)				
Core	3.1	<code>step_03_01_three_model_comparison.py</code>	Nested sampling (dynesty, <code>nlive</code> =500) for M0a_LCDM, M0b_EdS, M1 variants, M2_PureShear, M3_wCDM, M4_CPL; null injection	~90 min
Core	3.2	<code>step_03_02_independent_mcmc.py</code>	Independent MCMC convergence diagnostics	~1 s
Core	3.4	<code>step_03_04_cobaya_mcmc.py</code>	Joint SNe+CMB MCMC via Cobaya with TEP-CLASS v2.0	~2 min
Core	3.5	<code>step_03_05_analyze_cobaya.py</code>	Analyze Cobaya chains and produce parameter constraints	~1 s
Core	3.6	<code>step_03_06_cobaya_verbose.py</code>	Verbose Cobaya configuration and extended diagnostics	~2 min
Core	3.7	<code>step_03_07_likelihood_synthesis.py</code>	Synthesize likelihoods across independent and joint analyses	~1 s
Stage 4: Supernova Tests and Distance Duality (8 steps)				
SNe	4.1	<code>step_04_01_sn_time_dilation.py</code>	Test SN light-curve stretch factors against TEP time dilation	~1 s
SNe	4.2	<code>step_04_02_sn_tolman.py</code>	Tolman surface-brightness dimming test	~1 s
SNe	4.3	<code>step_04_03_tolman_sb.py</code>	Surface-brightness Tolman scaling with compiled catalog	~1 s
DDR	4.4	<code>step_04_04_distance_duality.py</code>	Distance-duality relation: BAO constraints vs TEP prediction	~1 s
DDR	4.5	<code>step_04_05_ddr_threeway.py</code>	Three-way probe comparison: BAO, SZ, SGL	~1 s
DDR	4.6	<code>step_04_06_screening_fit.py</code>	Parametric screening model fit to probe-dependent DDR	~2 s
DDR	4.7	<code>step_04_07_highz_ddr.py</code>	High-redshift Lyman-alpha DDR test (DESI, eBOSS)	~1 s
DDR	4.8	<code>step_04_08_environment_residuals.py</code>	Environmental residual analysis for probe-dependent screening	~1 s
Stage 5: CMB and Big Bang Nucleosynthesis (7 steps)				
CMB	5.1	<code>step_05_01_cmb_blackbody.py</code>	Verify TEP preserves CMB blackbody spectrum (FIRAS)	~1 s
CMB	5.3	<code>step_05_03_cmb_boltzmann.py</code>	TEP Boltzmann integration via patched CLASS	~1 s

Stage	Step	Script	Description	Runtime
CMB	5.4	step_05_04_cmb_spectra.py	Generate and compare TT/TE/EE power spectra	~1 s
CMB	5.5	step_05_05_cmb_consistency.py	CMB acoustic-scale consistency check	~1 s
BBN	5.6	step_05_06_bbn_registry.py	Compile observational BBN abundance registry	~1 s
BBN	5.7	step_05_07_bbn_preservation.py	Cross-validate TEP and LCDM BBN predictions	~1 s
CMB	5.8	step_05_08_cmb_acoustic.py	Acoustic-scale parameter comparison (Planck)	~1 s
Stage 6: BAO and Structure Growth (5 steps)				
BAO	6.1	step_06_01_bao_projection.py	BAO ruler projection in TEP geometry	~1 s
BAO	6.2	step_06_02_bao_likelihood.py	BAO likelihood module integration	~7 s
Growth	6.3	step_06_03_growth_solver.py	TEP-CLASS v2.0 growth equation solver	~1 s
Growth	6.4	step_06_04_growth_validation.py	Validate growth factors against LCDM baseline	~1 s
Growth	6.5	step_06_05_growth_rsd.py	Redshift-space distortion comparison (f sigma_8)	~2 s
Stage 7: Forecasts and Future Tests (7 steps)				
Future	7.1	step_07_01_mixed_forecast.py	Forecast for mixed TEP-LCDM parameter recovery	~1 s
Future	7.2	step_07_02_redshift_drift.py	Redshift-drift forecast and discriminating power	~1 s
Future	7.3	step_07_03_jwst_test.py	JWST high-z supernova feasibility test	~1 s
Future	7.4	step_07_04_gw_sirens.py	Gravitational-wave standard siren forecast	~1 s
Future	7.5	step_07_05_weak_lensing_plan.py	Weak-lensing survey plan for TEP discrimination	~1 s
Future	7.6	step_07_06_weak_lensing.py	Weak-lensing shear correlation analysis	~1 s
Future	7.7	step_07_07_blind_injection.py	Blind injection validation protocol	~1 s
Stage 8: Falsification, Audit, and Summary (8 steps)				
Audit	8.1	step_08_01_expansion_falsifier.py	Expansion falsifier: distance duality and Tolman residuals	~1 s
Audit	8.2	step_08_02_comparison_stats.py	Cross-model comparison statistics	~1 s
Audit	8.3	step_08_03_sensitivity_analysis.py	Prior and parameter sensitivity analysis	~1 s
Audit	8.4	step_08_04_evidence_matrix.py	Compile explanatory evidence matrix	~1 s
Audit	8.5	step_08_05_gate_registry.py	Claim gate registry and status check	~1 s
Audit	8.6	step_08_06_claim_audit.py	Automated claim consistency audit	~1 s
Audit	8.7	step_08_07_final_summary.py	Global evidence synthesis and summary	~1 s
Audit	8.8	step_08_08_diagnostic_plots.py	Publication-quality diagnostic and residual plots	~2 s

Total Runtime Summary

The total runtime is dominated by Stage 3.1 (nested sampling). Runtimes scale approximately linearly with `nlive` and number of CPU cores.

Component	Steps	Runtime
Data Acquisition (Stage 1)	8	~20 s
Theory and Transport (Stage 2)	3	~5 s
Model Comparison and MCMC (Stage 3)	6	~95 min
SNe Tests and DDR (Stage 4)	8	~10 s
CMB and BBN (Stage 5)	7	~8 s
BAO and Growth (Stage 6)	5	~12 s
Forecasts and Future Tests (Stage 7)	7	~7 s
Falsification and Audit (Stage 8)	8	~7 s
Total	52	~95 min (~1.6 h)

Reproduction Instructions

Quick Start (Full Reproduction)

```
# 1. Clone repository
git clone https://github.com/matthewsmawfield/TEP-C0.git
cd TEP-C0

# 2. Install dependencies
pip install -r requirements.txt

# 3. Run full pipeline (generates all results and figures)
python scripts/run_pipeline.py

# 4. Results will be in:
#   - results/outputs/   (JSON/CSV data)
#   - results/figures/   (PNG/PDF plots)
#   - logs/              (Detailed execution logs)
```

Command-Line Options

The pipeline supports selective execution for faster testing:

```
# Core statistical analysis only (skips long nested sampling)
python scripts/run_pipeline.py --core

# Resume from existing results (skip completed steps)
python scripts/run_pipeline.py --resume

# Run specific steps with automatic dependency resolution
python scripts/run_pipeline.py --steps step_04_04_distance_duality
step_04_05_ddr_threeway
```

System Requirements

Component	Minimum	Recommended	Tested On
CPU	4 cores	8+ cores	Apple M4 Pro (14-core)
RAM	8 GB	16 GB	24 GB (M4 Pro)
Storage	2 GB	5 GB	NVMe SSD
Runtime (full)	~4 h (4 cores)	~1.5 h (8+ cores)	~95 min (M4 Pro)
Runtime (--core)	~1 min	~30 s	~20 s

Key Analysis Outputs

- `results/outputs/step_03_01_three_model_comparison.json` — Nested sampling posteriors and evidence for all models (M0a_LCDM, M0b_EdS, M1 variants, M2_PureShear, M3_wCDM, M4_CPL)
- `results/outputs/step_03_04_cobaya_mcmc.1.txt` — Cobaya MCMC chain for joint SNe+CMB analysis
- `results/outputs/step_04_04_distance_duality.json` — DDR weighted mean and deviation from unity
- `results/outputs/step_04_05_ddr_threeway.json` — Three-way BAO/SZ/SGL probe comparison
- `results/outputs/step_05_07_bbn_preservation.json` — TEP vs LCDM light-element abundance cross-validation
- `results/outputs/step_06_04_growth_validation.json` — Growth factor and σ_8 consistency check
- `results/outputs/step_08_04_evidence_matrix.json` — Explanatory evidence matrix across all observables
- `results/outputs/step_08_06_claim_audit.json` — Automated claim consistency audit report

Log Files

Each step produces detailed logs with timestamps, SHA-256 checksums, and execution status:

- `logs/step_*.log` — Individual step logs (52 files, one per step)

- `logs/verbose/` — Verbose Cobaya and nested sampling logs

Software Dependencies

Package	Version	Purpose
Python	3.10+	Language runtime
NumPy	1.24+	Numerical computing
SciPy	1.10+	Statistical functions, nested sampling
Pandas	2.0+	Data manipulation
Matplotlib	3.7+	Visualization
emcee	3.1+	Ensemble MCMC sampling
dynesty	2.1+	Nested sampling for Bayesian evidence
Cobaya	3.6+	Joint MCMC with Planck likelihoods
classy (CLASS)	3.2+	CMB Boltzmann solver (patched for TEP)

All dependencies are specified in `requirements.txt`. External dependencies (patched CLASS, AlterBBN) are included in the `external/` directory.